Dong 2020 Prostate Cancer — analysis summary

GO / NO-GO GATES

Gate 1 — Integration & continuity **GO**

Malignant cells from all 4 patients (9,698; ≥100-cell floor, no sample dropped) form one continuous manifold — 0 detached islands, single connected component; every transcriptional lobe contains all 4 patients. Judged on the embedding (samples.pdf), not the composition-capped iLISI (1.78/4, one patient ≈49% of cells).

Gate 2 — Embryonic module geography **GO (non-branching)**

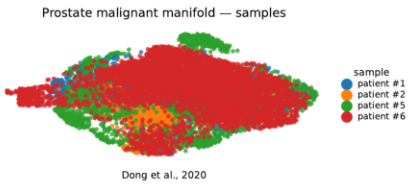
Single developmental axis: Gast & pSom co-vary (smoothed r=+0.90; joint-high enrichment 2.4 — an overlap, not competing arms) and jointly oppose neural-crest (pSom↔Crest r=−0.53). Along pseudotime Gast/pSom decline (rho −0.58/−0.71), Crest rises to a terminal end (rho +0.79, Crest-high decile pt 0.98). Non-branching — passes.

Gate 3 — Hub cross-repression **NA**

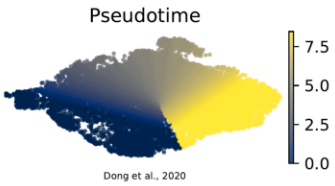
No bifurcation (fold-free single curve, 2 tips / 0 forks). The branching-only cross-repression test does not apply; correlation & modules notebooks are not produced (§6 renumbering: expression = 04).

STEP FIGURES

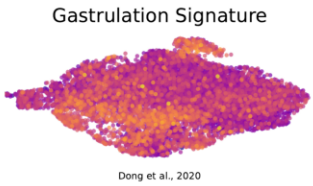
Sample integration (Gate 1)



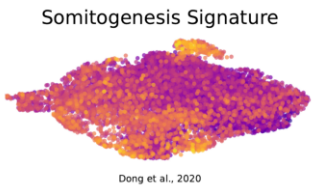
Pseudotime trajectory



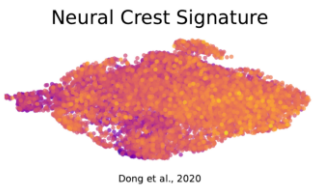
Gastrulation



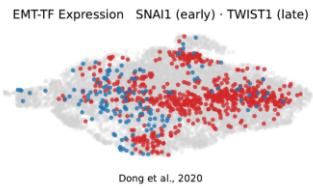
Somitogenesis



Neural crest



EMT-TF (SNAI1→TWIST1)



Dataset. Prostate adenocarcinoma, 10x (Dong et al. 2020); 9,698 malignant cells across 4 patients (publication cell_type calls). Non-branching embryonic axis.

Verdict. The embryonic program reproduces as a single monotone gradient — a Gast/pSom-high (overlap) origin flowing to an inverse neural-crest terminus — the collapsed-bifurcation (Linear-pSom/Crest) case, distinct from Puram's hub-and-spoke. G1 GO / G2 GO / G3 NA.

Parameter choices

Malignant / threshold:	9,698 cells, 4 patients; ≥100 floor (no multiplicative cliff: gap ratios 2.03/1.40/1.83, top/runner-up 1.11<1.5) — all samples kept
Overdispersion / PCA:	1,670 overdispersed (scFates); 50 PC → 20 for kNN
Integration:	Harmony on sample, theta=10 (strong patient-specific batch, one patient ≈49%); spread=1.3
Island removal:	radius graph 6× median NN → 0 cells removed (one component)
Compartments:	Leiden res=0.15 → Body (low-cycling, 8,696) + Cycling (1,002, 94% cyc); whole manifold kept
Cell-cycle confound:	axis is cell-cycle-free: modules 0–2/~90 CC-gene overlap; anti-corr survives in 7,988 non-cycling cells; pt↔cyc rho −0.24 (caveat, not scoping)
Trajectory:	scFates curve Nodes=20 epg_lambda=0.3 epg_mu=100 (X_umap); 2 tips 0 forks, fold self-approach 2.59 (stable N=15/20/30)
Root:	tip 2 (Gast/pSom-high, Crest-low, argmax); pseudotime origin→Crest terminus
Orientation:	rigid rotation −170.4° (distances preserved); corr(x_lin, pt)=+0.85
EMT-TFs (derived):	SNAI1 (root, 6.0%) ↔ TWIST1 (terminus, 26.8%) by rho-vs-pt; end-separation dpt=0.37